PRINCIPAL COMPONENT ANALYSIS (PCA)

Principal Component Analysis (PCA) is a **dimensionality reduction** technique used to reduce the number of variables in a dataset while preserving as much variability (information) as possible.

In short, PCA used to analyze multivariate data.
Examples of multivariate data are

a. Patient medical record: One person might have:
Age, Blood pressure, Heart rate, Cholesterol level, Glucose level

b. Weather data (for one day):
Temperature, Humidity, Wind speed, Rainfall

c. Consider the following **multivariate data** with 3 variables: Height, Weight, and Age of students .

Student	Height (cm)	Weight (kg)	Age (years)
A	170	65	20
B	160	58	22
C	180	75	21

Each student here is an **observation**, and each column is a **variable**.

What is PRINCIPAL OF COMPONENT ANALYSIS?

Principal Component Analysis (PCA) is a **dimensionality reduction** technique used to reduce the number of variables in a dataset while preserving as much preserving as much variability (information) as possible.

If you have 2D data:

- PCA finds a new axis/direction where data varies the most.

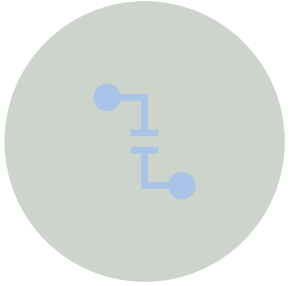
OBJECTIVES OF PCA



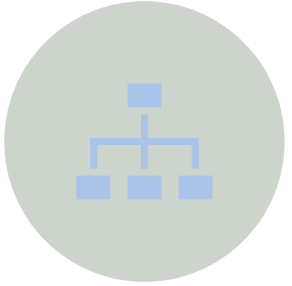
Simplify the complexity in high-dimensional data.



Identify patterns and highlight similarities/differences.



Reduce noise and redundancy in data.



Improve efficiency of other algorithms (e.g., clustering, classification).

Key Concepts



Variance: Measure of spread. PCA keeps components with the most variance.



Covariance Matrix: Captures how variables vary together.



Eigenvectors: Directions of maximum variance (principal components).



Eigenvalues: Magnitude of the variance in those directions.

Working rule for pca

1. **Standardize the data**
   (mean = 0, standard deviation = 1)
2. **Compute the covariance matrix**
3. **Calculate eigenvalues and eigenvectors** of the covariance matrix.
4. **Sort eigenvectors** by decreasing eigenvalues — these are your principal components.
5. **Project the data** onto the top *k* eigenvectors (components).

EXAMPLE

Find principal components of the given data, where each row is a data point, each column a feature/property (x and y).

$$X = \begin{bmatrix} 2.5 & 2.4 \\ 0.5 & 0.7 \\ 2.2 & 2.9 \end{bmatrix}$$

SOLUTION:

Step 1: Mean Centering the Data

First, subtract the mean of each column (feature):

Column means:

$$\mu_x = \frac{2.5 + 0.5 + 2.2}{3} = 1.733, \qquad \mu_y = \frac{2.4 + 0.7 + 2.9}{3} = 2.0$$

$$X - \mu = \begin{bmatrix} 2.5 - 1.733 & 2.4 - 2.0 \\ 0.5 - 1.733 & 0.7 - 2.0 \\ 2.2 - 1.733 & 2.9 - 2.0 \end{bmatrix} = \begin{bmatrix} 0.767 & 0.4 \\ -1.233 & -1.3 \\ 0.467 & 0.9 \end{bmatrix}$$

Step 2: Covariance Matrix

$$\text{Cov}(X) = \frac{1}{n-1} X^T X$$

Computing the covariance matrix

$$\text{Cov}(X) = \frac{1}{2} \begin{bmatrix} 0.767 & -1.233 & 0.467 \\ 0.4 & -1.3 & 0.9 \end{bmatrix} \begin{bmatrix} 0.767 & 0.4 \\ -1.233 & -1.3 \\ 0.467 & 0.9 \end{bmatrix}$$

$$\text{Cov}(X) = \begin{bmatrix} 1.163 & 1.165 \\ 1.165 & 1.33 \end{bmatrix}$$

Step 3: Compute Eigenvalues and Eigenvectors

Solve the characteristic equation:

$$\det(\text{Cov} - \lambda I) = 0$$

The eigenvalues are

The largest eigenvalue is 2.414.

Find eigenvectors corresponding to only the largest eigenvalue

Step 4: Project Data onto Principal Component

Project centered data onto the eigenvector by multiplying the following.

$$X_{\text{projected}} = X_{\text{centered}} \cdot \vec{v}_1$$

We have transformed our 2D data into 1D along the axis of maximum variance.

PRACTICE PROBLEMS

The following table lists the weights and heights of five boys:

Boy	#1	#2	#3	#4	#5
Weight (lb)	120	125	125	135	145
Height (in.)	61	60	64	68	72

1. Find the covariance matrix for the data.
2. Make a principal component analysis of the data to find a single *size index* that explains most of the variation in the data.

E1): Weather data given below pertains to Raipur district (M.P.) from 1970 to 1986 for kharif crop season from 21st May to 7th October. The weather variables are average minimum temperature (x_1), average relative humidity at 8 hrs (x_2), average relative humidity at 14 hrs (x_3) and total rainfall in cm (x_4).

S.No.	x_1	x_2	x_3	x_4
1	25	86	66	186.49
2	24.9	84	66	124.34
3	25.4	77	55	98.79
4	24.4	82	62	118.88
5	22.9	79	53	71.88
6	7.7	86	60	111.96
7	25.1	82	58	99.74
8	24.9	83	63	115.2
9	24.9	82	63	100.6
10	24.9	78	56	62.38
11	24.3	85	67	154.4
12	24.6	79	61	112.71
13	24.3	81	58	79.63
14	24.6	81	61	125.59
15	24.1	85	64	99.87
16	24.5	84	63	143.56
17	24	81	61	114.97
Mean	**23.56**	**82.06**	**61**	**112.97**
S.D.	**4.13**	**2.75**	**3.97**	**30.06**

Find the principal components using variance covariance matrix and correlation matrix.

E1) The problem is to find first few principal components, which may account for most of the variation in the original data.

I. Identification of important principal components through covariance matrix of the original variables.

The variables $\mathbf{X}' = (x_1, x_2, x_3, x_4)$ have the covariance matrix as

$$\mathbf{\Sigma} = \begin{bmatrix} 17.02 & -4.12 & 1.54 & 5.14 \\ & 7.56 & 8.50 & 54.82 \\ & & 15.75 & 92.95 \\ & & & 903.87 \end{bmatrix}.$$

The eigenvalue-eigenvector pairs of $\mathbf{\Sigma}$ are

$\lambda_1 = 916.902 \quad \mathbf{a}_1' = (0.006, \quad 0.061, \quad 0.103, \quad 0.993)$

$\lambda_2 = 18.375 \quad \mathbf{a}_2' = (0.955, \quad -0.296, \quad 0.011, \quad 0.012)$

$\lambda_3 = 7.87 \quad \mathbf{a}_3' = (0.141, \quad 0.485, \quad 0.855, \quad -0.119)$

$\lambda_4 = 1.056 \quad \mathbf{a}_4' = (0.260, \quad 0.820, \quad -0.509, \quad .001).$

The principal component are $\mathbf{Z}_i = \mathbf{a}_i' \mathbf{x}, i = 1,2,3,4.$